



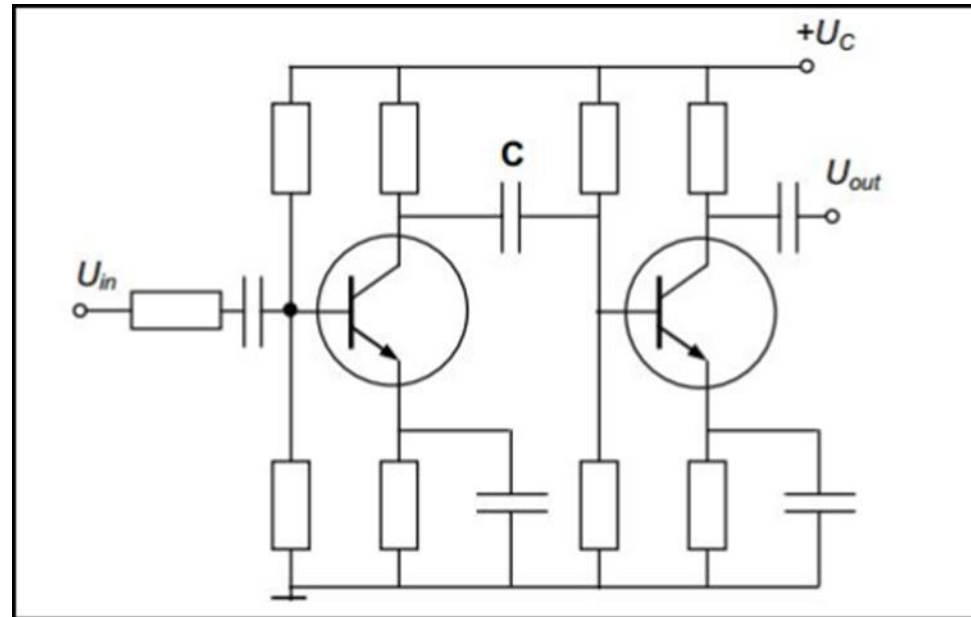
Penguat Tegangan

PDS

<https://www.youtube.com/@AmelOline>

<https://github.com/amelcharolinesgn2/DE/upload>

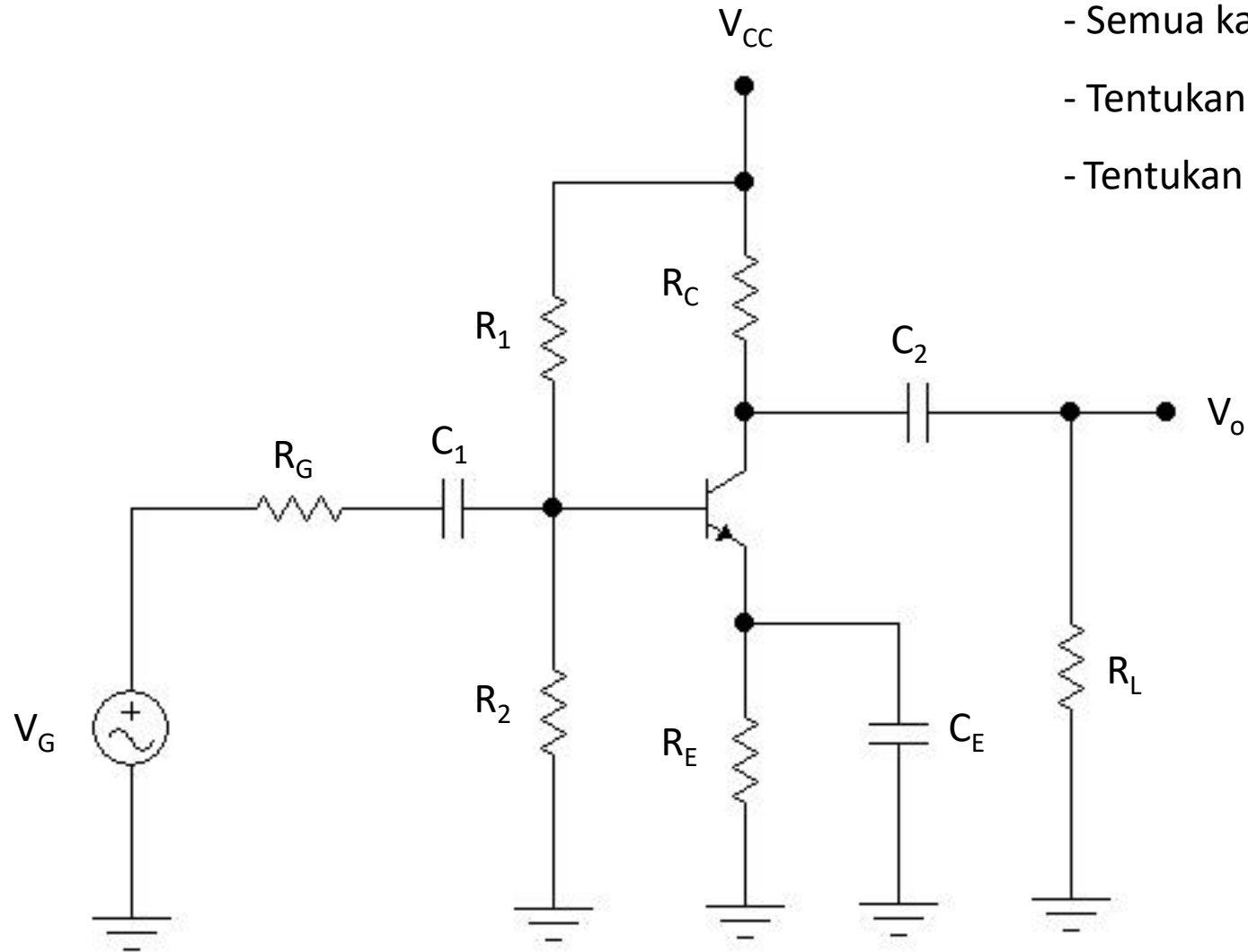
Transistor amplifiers

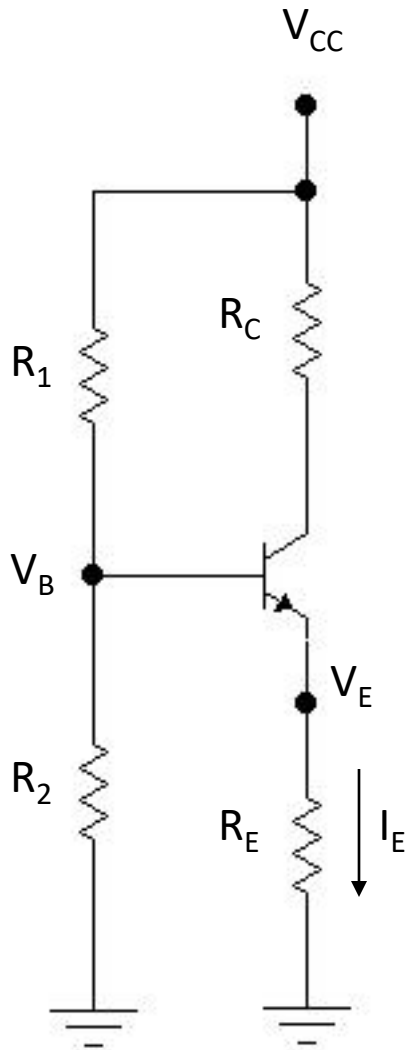


PENGUAT COMMON EMITTER

Analisis DC ($f = 0$)

- Semua sumber AC = 0
- Semua kapasitor open
- Tentukan arus emiter I_E
- Tentukan tahanan dalam r_e'





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$$V_B = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$V_E = V_B - V_{BE} = V_B - 0,7$$

$$I_E = \frac{V_E}{R_E}$$

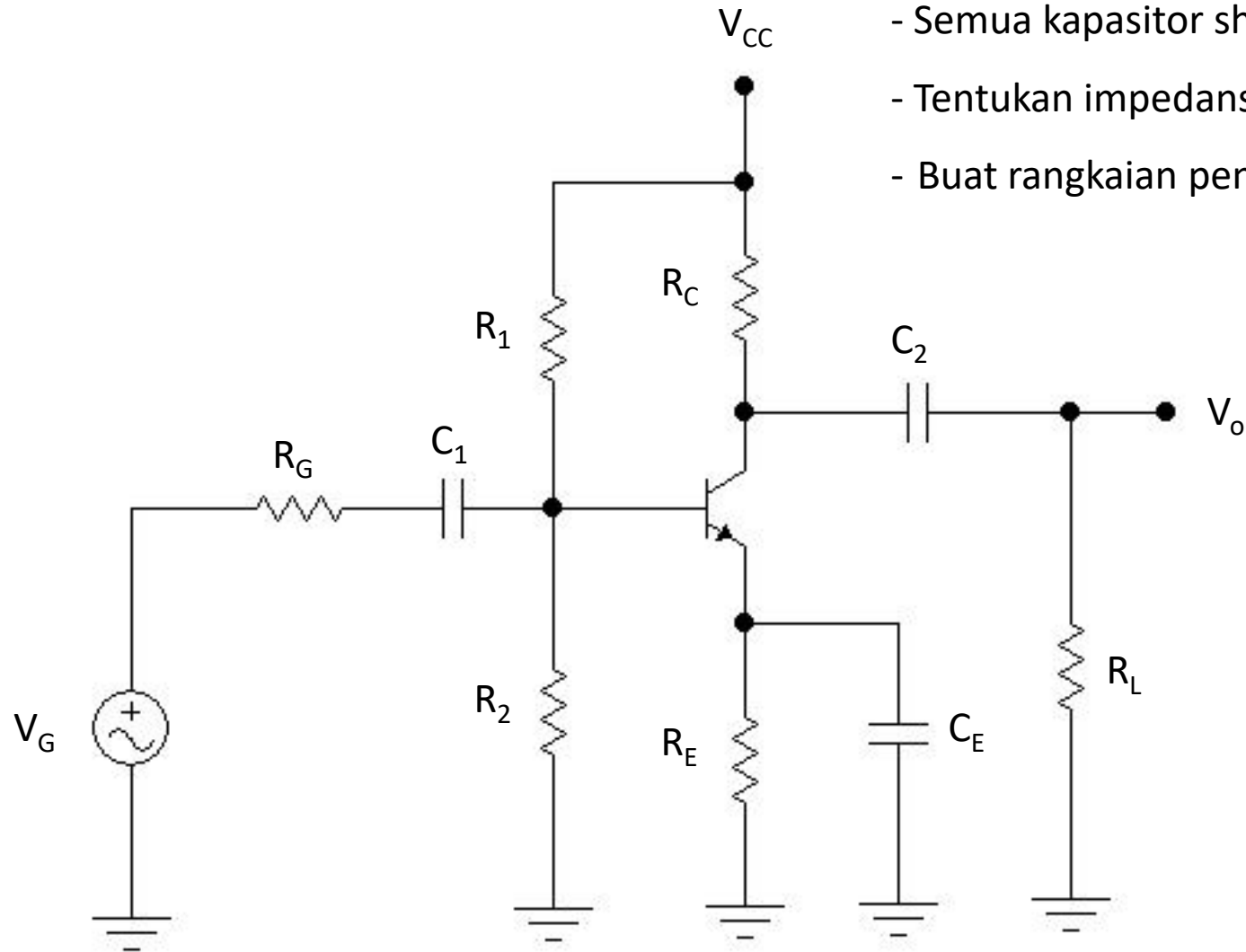
$$r_e' = \frac{25\text{mV}}{I_E} \frac{T + 273}{298}$$

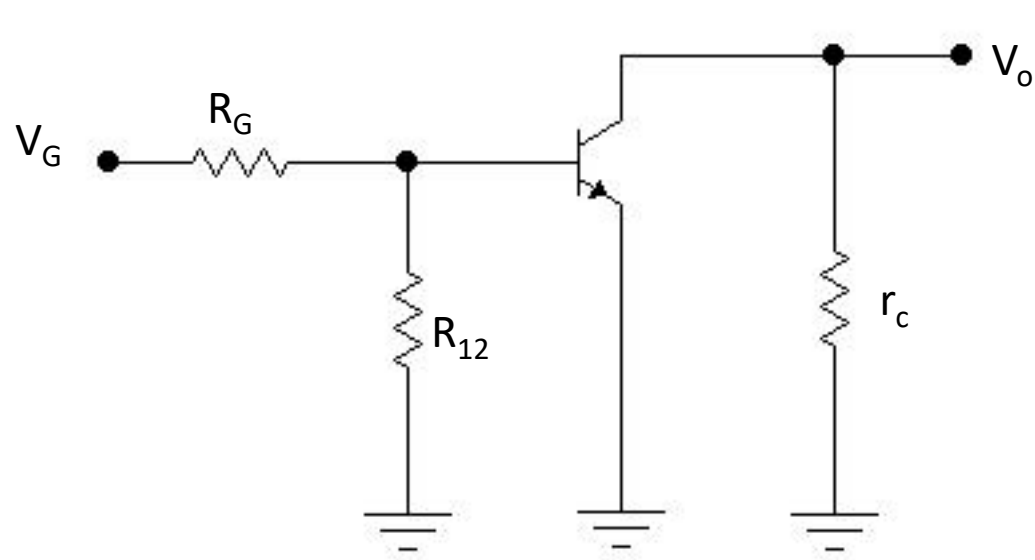
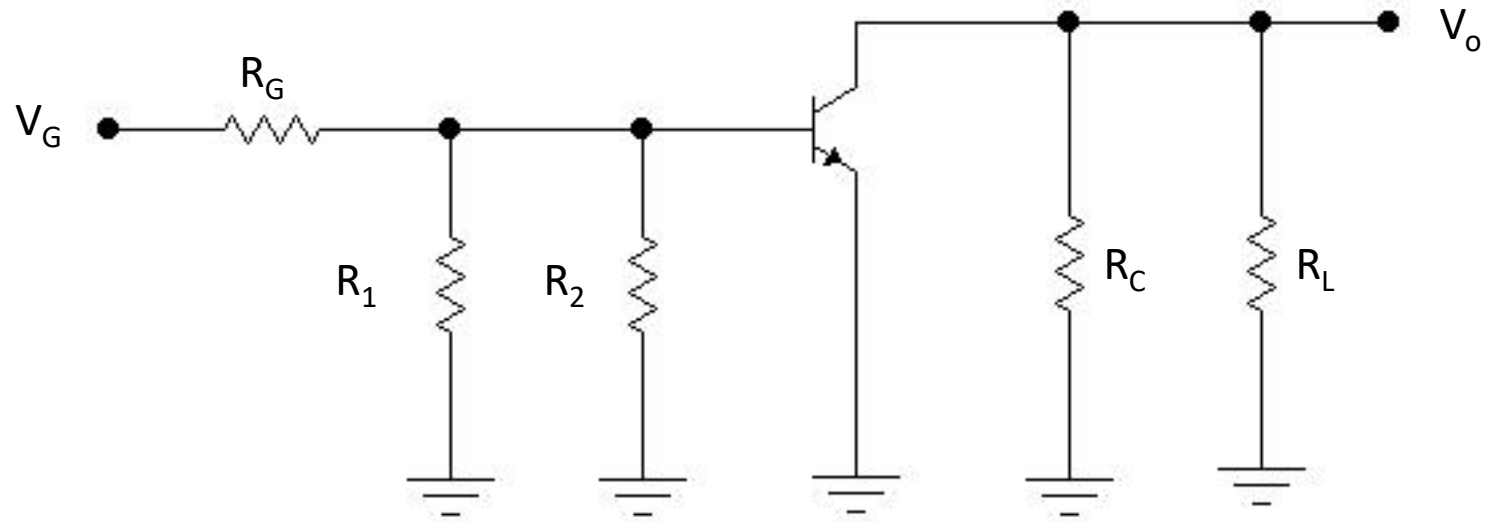
$$25^\circ\text{C} \rightarrow T = 273 + 25 = 298\text{K}$$

$$r_e' = \frac{25\text{mV}}{I_E}$$

Analisis ac ($f = \infty$)

- Semua sumber DC = 0
- Semua kapasitor short
- Tentukan impedansi input basis Z_{inbase}
- Buat rangkaian pengganti ac



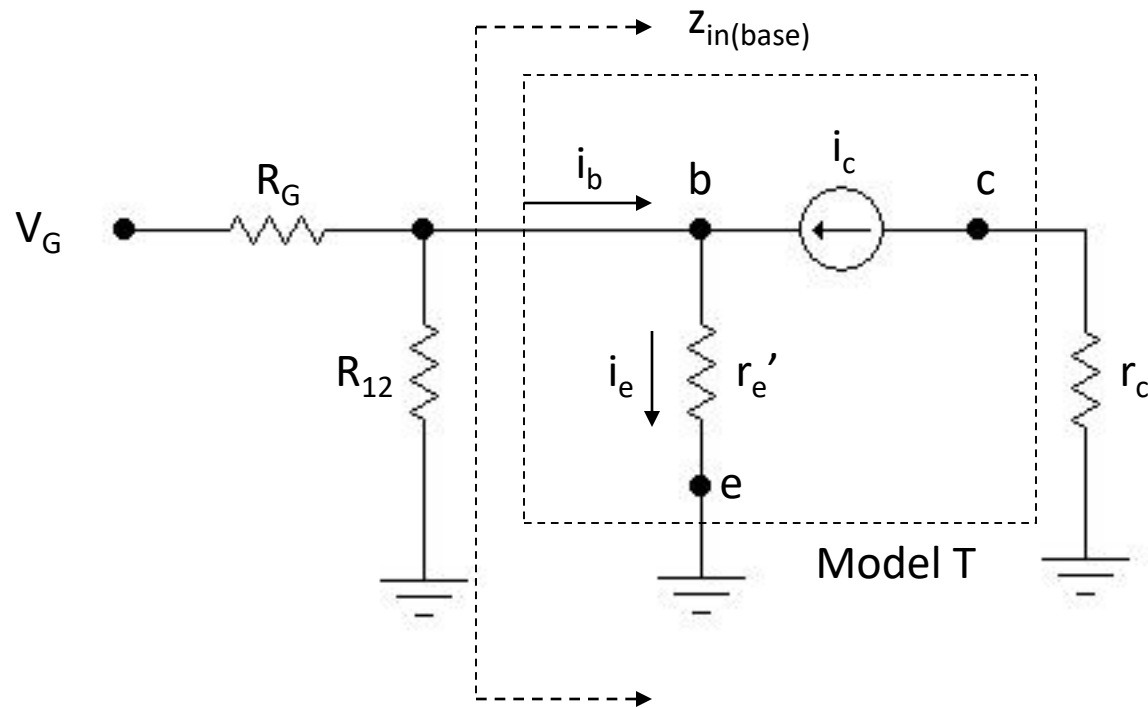


$$R_{12} = R_1 // R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

$$r_c = R_C // R_L = \frac{R_C R_L}{R_C + R_L}$$

↓
Tahanan kolektor ac

RANGKAIAN PENGGANTI TRANSISTOR



$$V_c = r_c i_c \quad i_e = \frac{V_b}{r_e'}$$

$$\beta = \frac{i_c}{i_b} \quad i_e = i_b + i_c$$

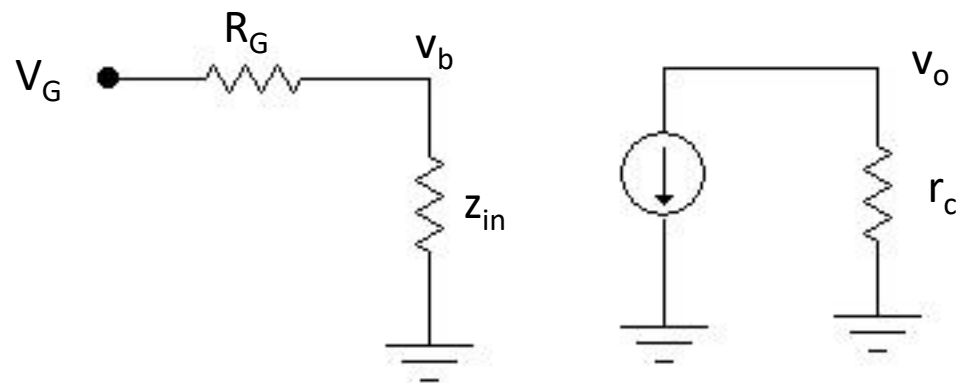
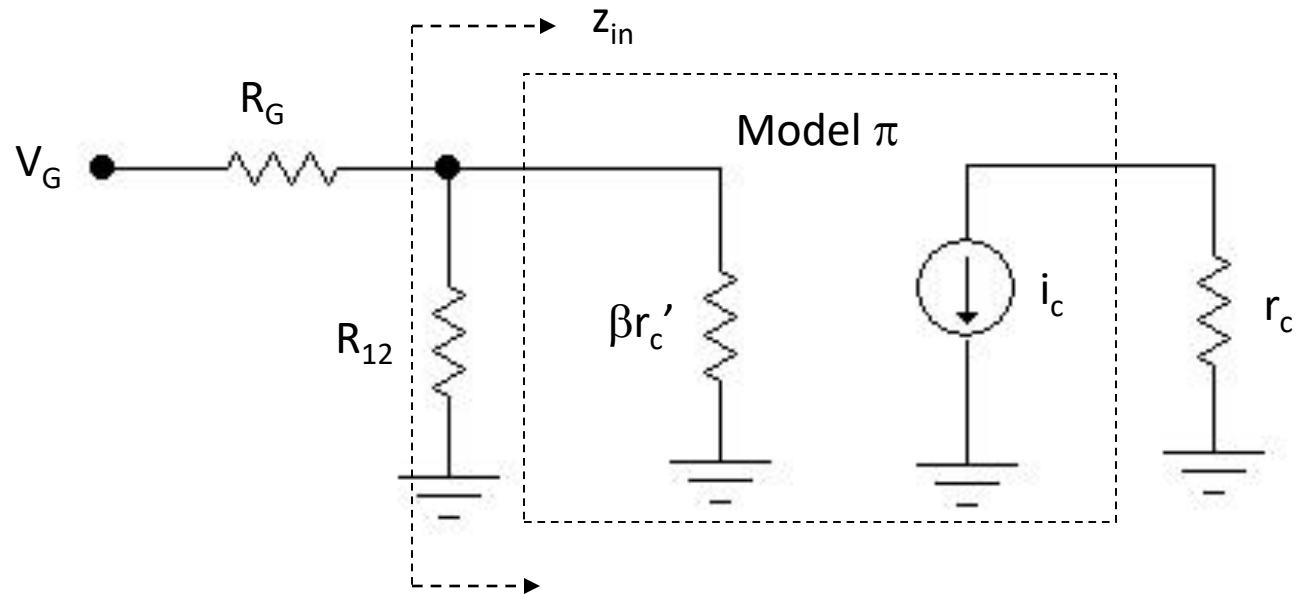
$$A_v = \frac{V_c}{V_b} = \frac{r_c i_c}{r_e' i_e}$$

$$\beta \gg 1 \rightarrow i_e \approx i_c$$

$$A_v = \frac{r_c}{r_e'}$$

$$Z_{in(base)} = \frac{V_b}{i_b} = \frac{i_e r_e'}{i_b}$$

$$\beta \gg 1 \rightarrow Z_{in(base)} = \beta r_e'$$



$$Z_{in} = R_1 // R_2 // \beta r'_e$$

$$V_b = V_{in} = \frac{Z_{in}}{Z_{in} + R_G} V_G$$

$$V_o = A_v V_{in}$$

$$= \frac{r_c}{r'_e} \frac{Z_{in}}{Z_{in} + R_G} V_G$$

Contoh soal 4.1

a). Tentukan tegangan output dari penguat CE di bawah ini

b). Berapa outputnya bila temperaturnya naik menjadi 150°C

$$V_B = \frac{2,2K}{2,2K + 10K} (10V) = 1,8V$$

$$V_E = 1,8 - 0,7 = 1,1V$$

$$I_E = \frac{1,1V}{1K\Omega} = 1,1mA$$

$$r'_e = \frac{25mV}{1,1mA} = 22,7\Omega$$

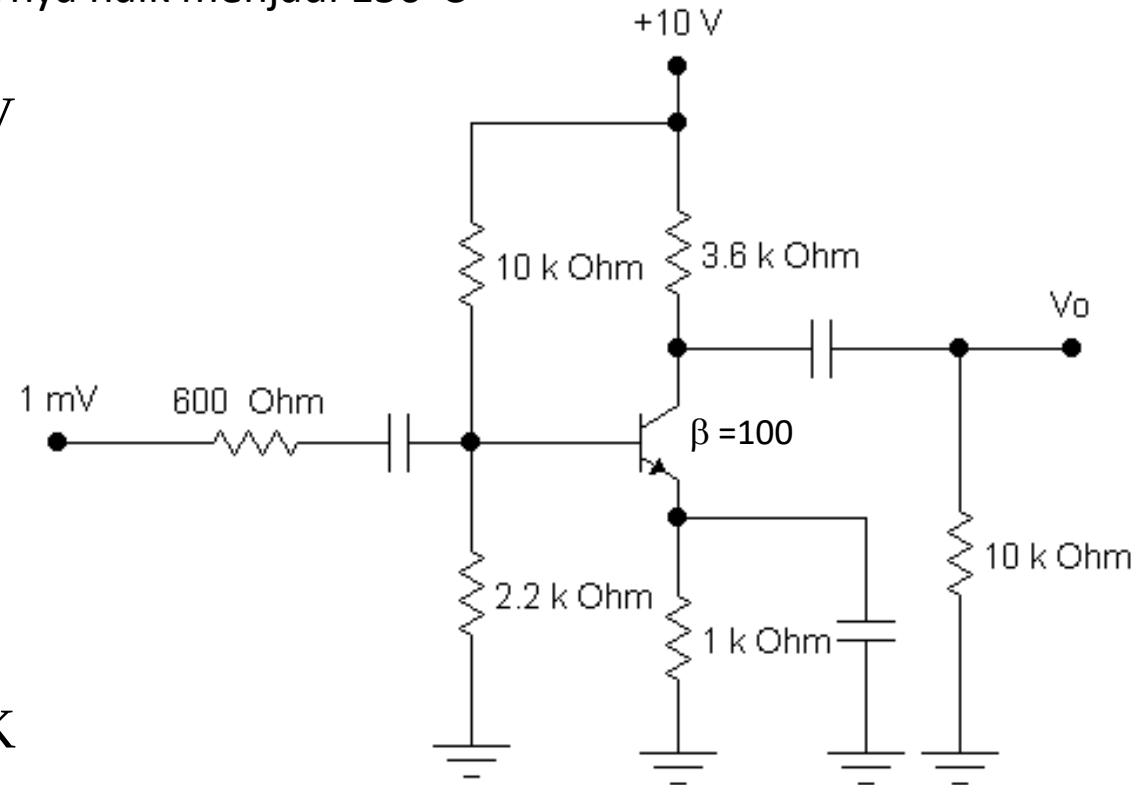
$$Z_{in(base)} = 100(22,7) = 2,27K\Omega$$

$$Z_{in} = 10K // 2,2K // 2,27K = 1K$$

$$V_{in} = \frac{1K}{1K + 600} (1mV) = 0,625mV$$

$$r_c = 3,6K // 10K = 2,65K\Omega \rightarrow A = \frac{2,65K}{22,7} = 117$$

$$V_o = (117)(0,625) = 72,9mV$$



$$r'_e = \frac{25\text{mV}}{1,1\text{mA}} \frac{273+150}{298} = 32,3\Omega$$

$$Z_{\text{in}(\text{base})} = 100(32,3) = 3,23\text{K}\Omega$$

$$Z_{\text{in}} = 10\text{K} // 2,2\text{K} // 3,23\text{K} = 1,16\text{K}$$

$$V_{\text{in}} = \frac{1,16\text{K}}{1,16\text{K} + 600} (1\text{mV}) = 0,659\text{mV}$$

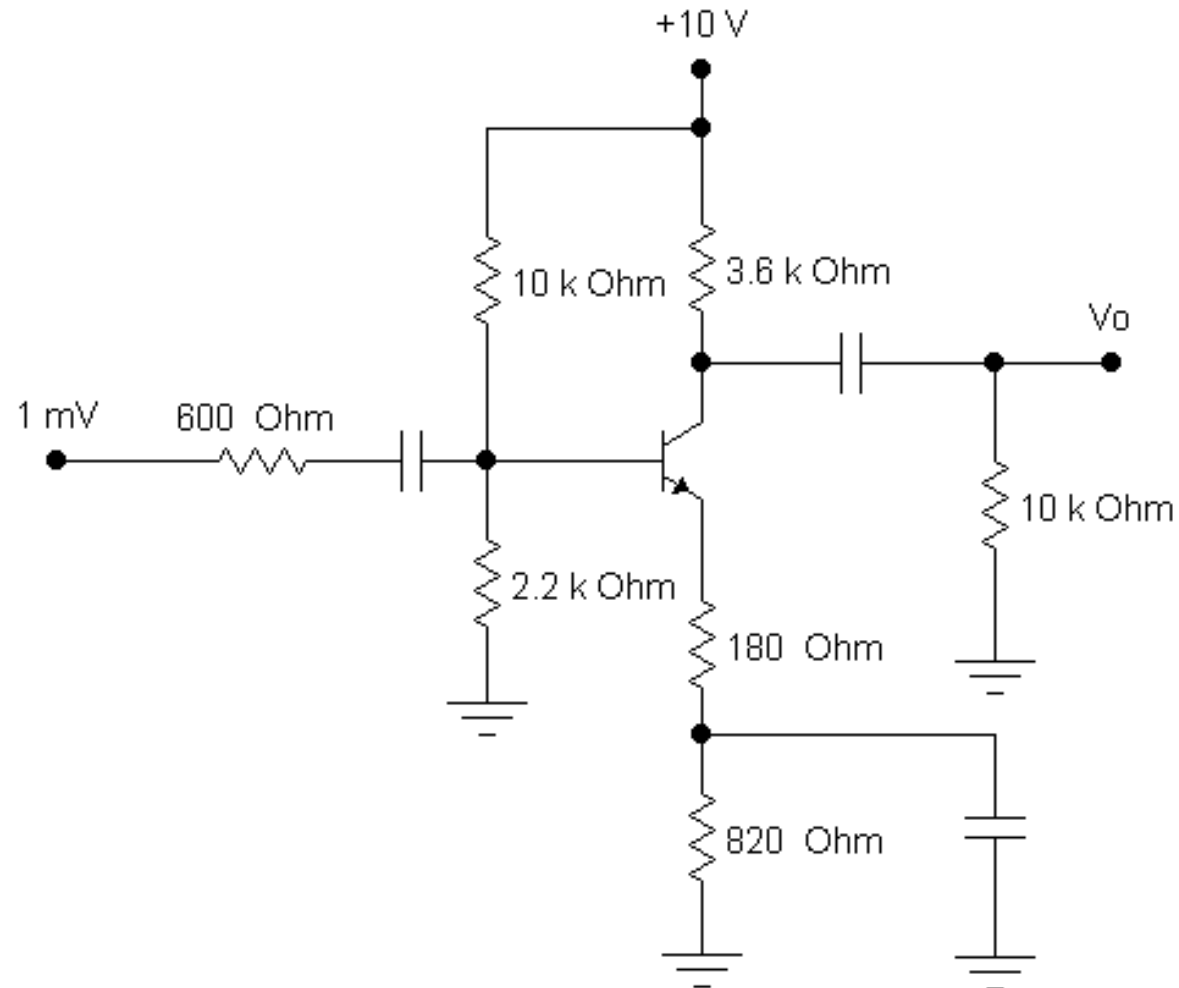
$$r_c = 3,6\text{K} // 10\text{K} = 2,65\text{K}\Omega \quad \rightarrow \quad A = \frac{2,65\text{K}}{32,3} = 54$$

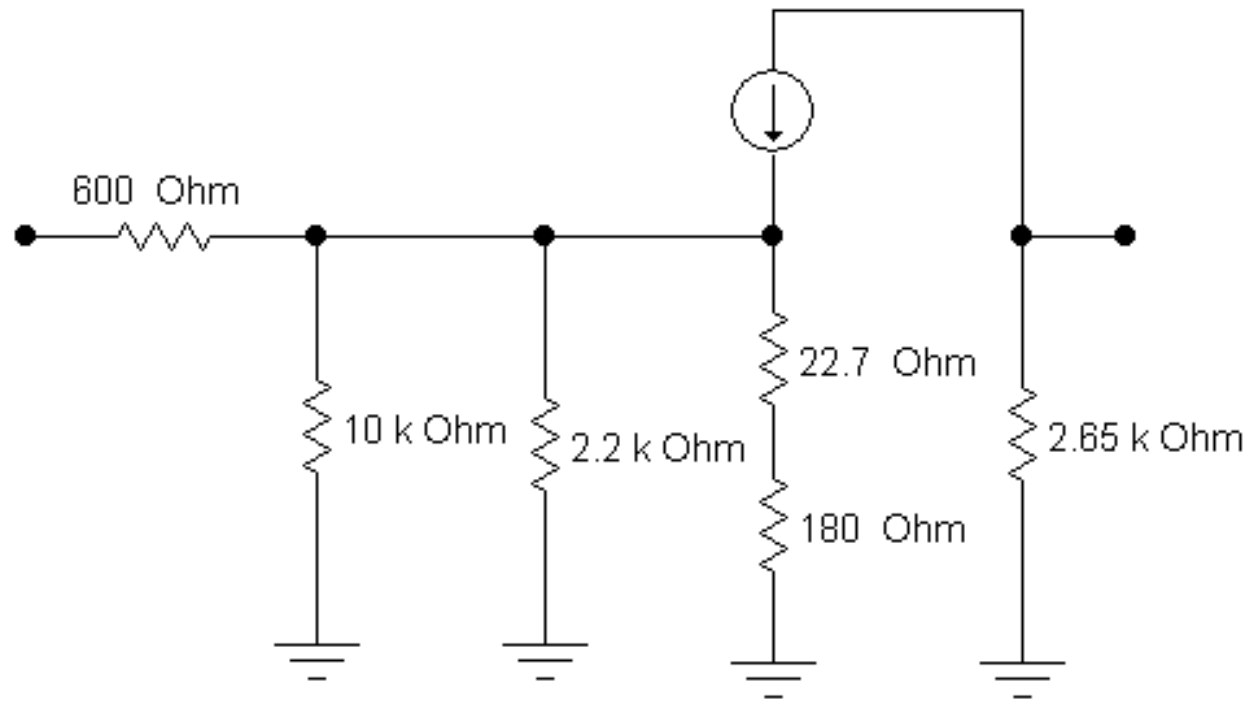
$$V_o = (54)(0,659) = 35,3\text{mV} \quad \rightarrow \quad \% \Delta V_o = 52\%$$

Contoh Soal 4.2

Rangkaian di bawah ini disebut sebagai *swamped amplifier*.

Hitung persentase perubahan outputnya bila temperatur berubah dari 25°C menjadi 150°C .





$$25^{\circ}\text{C} \rightarrow r'_e = 22,7\Omega \rightarrow z_{\text{in(base)}} = 100(22,7 + 180) = 20,27\text{K}\Omega$$

$$z_{\text{in}} = 10\text{K} // 2,2\text{K} // 20,27\text{K} = 1,65\text{K}\Omega$$

$$v_{\text{in}} = \frac{1,65\text{k}}{1,65\text{K} + 600} (1\text{mV}) = 0,733\text{mV}$$

$$A = \frac{2,65\text{K}}{22,7 + 180} = 13,1 \rightarrow V_o = 0,733(13,1) = 9,6\text{mV}$$

$$150^{\circ}\text{C} \rightarrow r_e' = 32,3\Omega \rightarrow z_{\text{in}(\text{base})} = 100(32,3 + 180) = 21,23\text{K}\Omega$$

$$z_{\text{in}} = 10\text{K} // 2,2\text{K} // 21,23\text{K} = 1,66\text{K}\Omega$$

$$v_{\text{in}} = \frac{1,66\text{k}}{1,66\text{K} + 600} (1\text{mV}) = 0,735\text{mV}$$

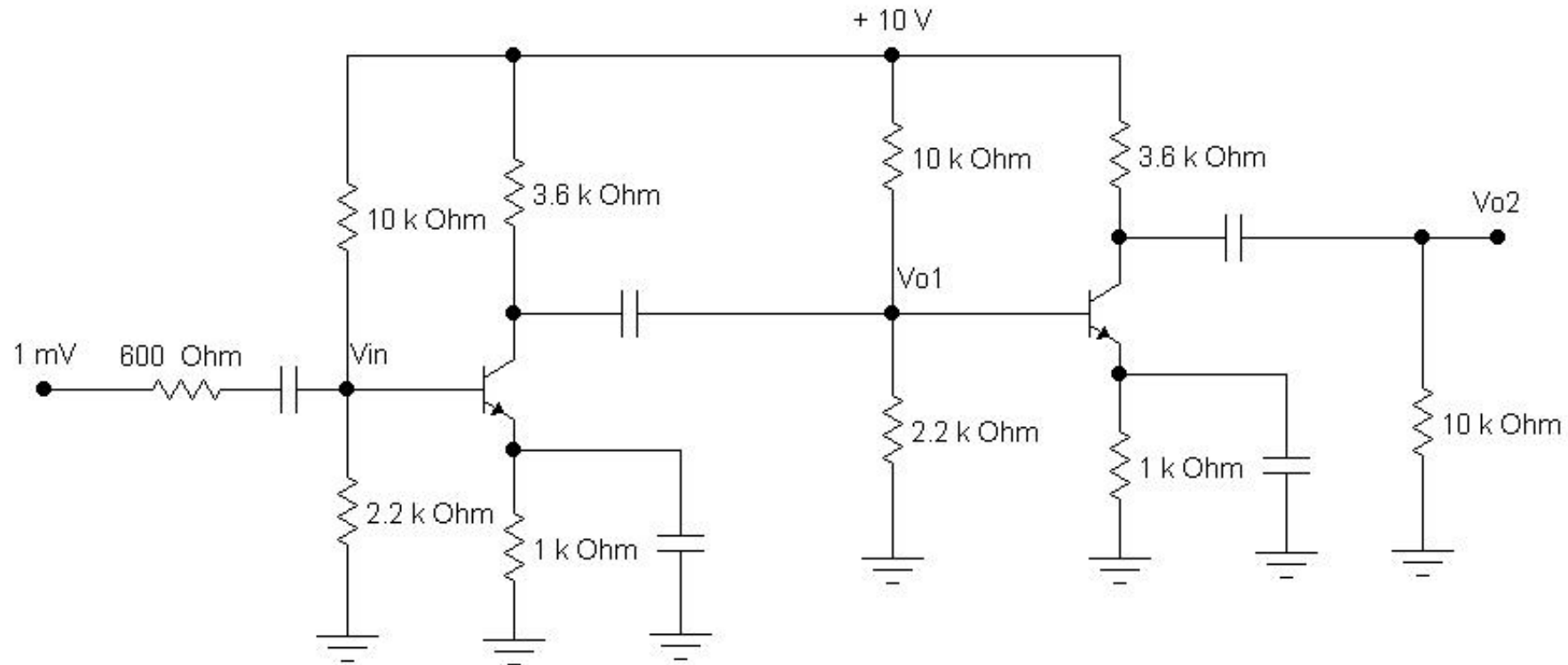
$$A = \frac{2,65\text{K}}{32,3 + 180} = 12,5 \rightarrow V_o = 0,735(12,5) = 9,2\text{mV}$$

$$\% \Delta V_o = \frac{9,6 - 9,2}{9,6} (100) = 4,2\%$$

Swamped amplifier lebih stabil tetapi penguatannya lebih kecil

Contoh Soal 4.3

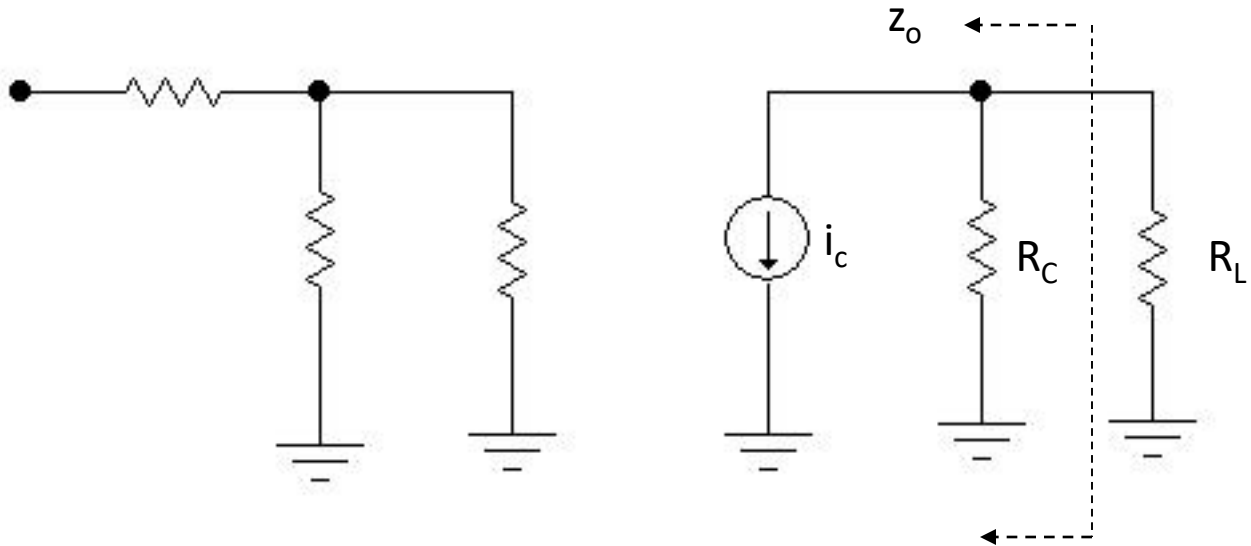
Untuk memperbesar tegangan output, pada rangkaian di bawah ini digunakan dua penguat CE yang dihubungkan secara seri (kaskade). Hitung tegangan outputnya.



$$A_2 = \frac{V_{o2}}{V_{o1}} = 117 \quad Z_{in2} = Z_{L1} = 1K\Omega \quad r_{c1} = 3,6K // 1K = 783\Omega$$

$$A_1 = \frac{783}{22,7} = 34,5 \quad V_{in} = 0,625mV \rightarrow V_{o1} = (34,5)(0,625) = 21,6mV$$

$$V_{o2} = (117)(21,6) = 2527mV \neq (117)(117) = 13689mV$$

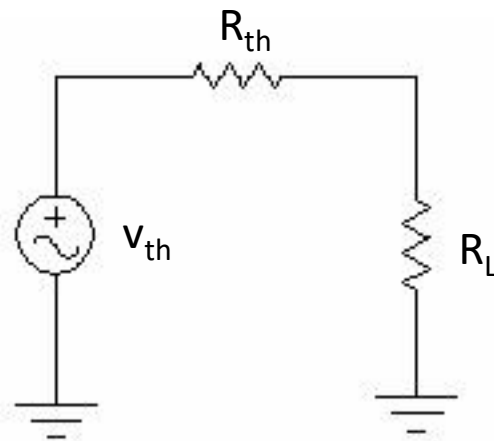


Thevenin :

$$Z_o = R_{th} = R_C$$

$$V_o = \frac{R_L}{R_L + Z_o} V_{th}$$

$$V_{th} = i_c R_C = \frac{V_{in}}{r'_e} R_C \rightarrow A_{th} = \frac{V_{th}}{V_{in}} = \frac{R_C}{r'_e}$$



PENGUAT COMMON COLLECTOR

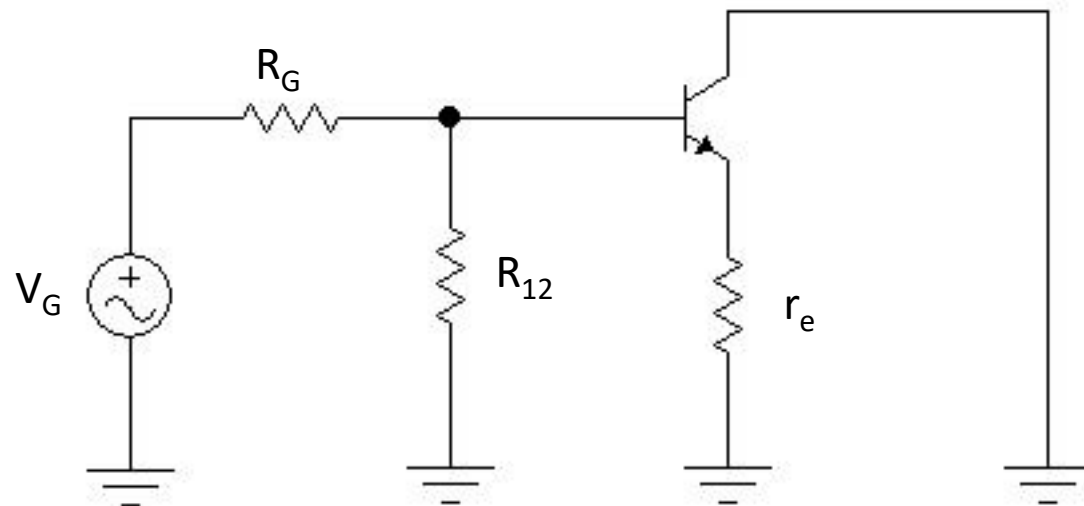
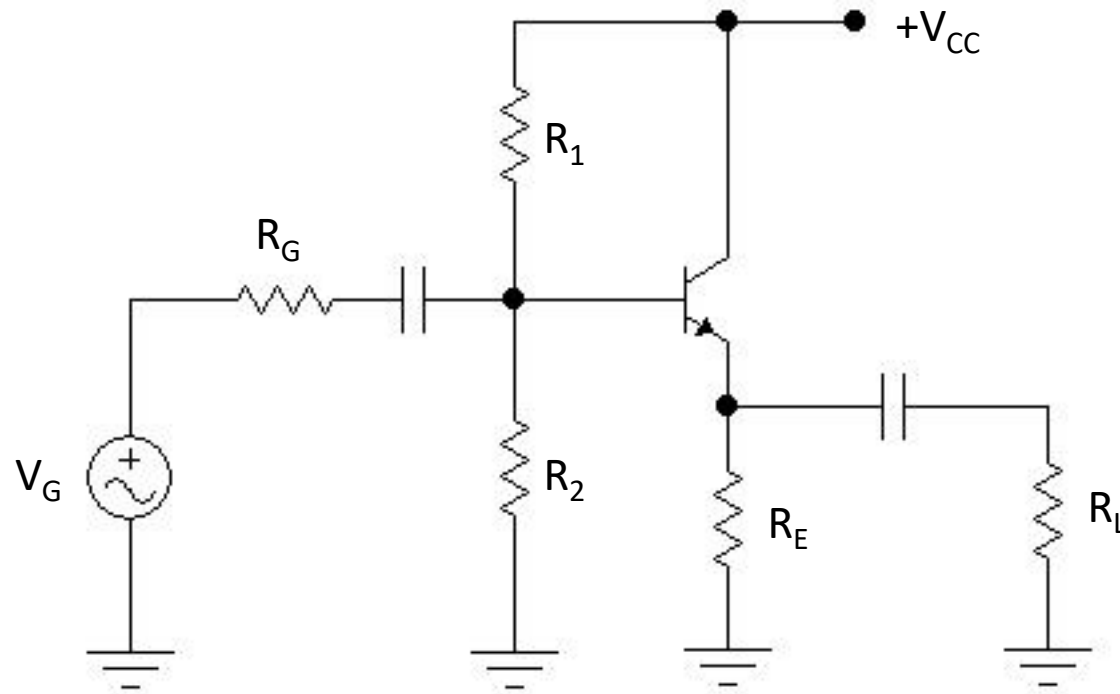
Analisis DC :

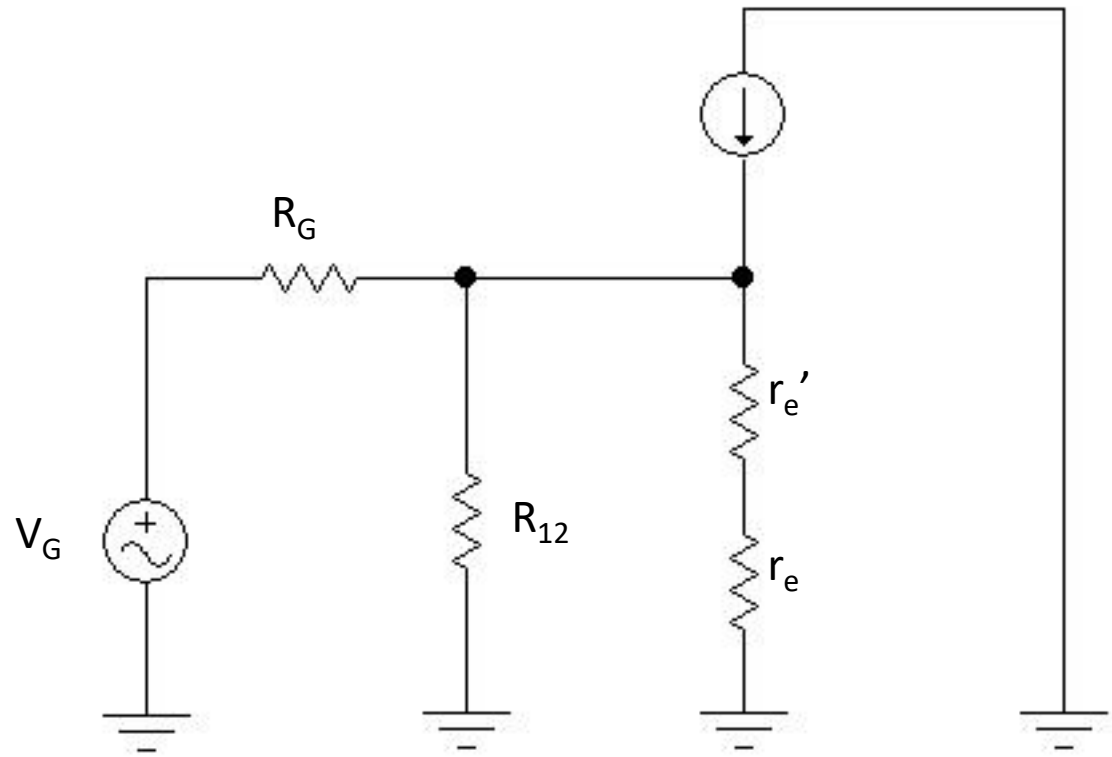
- Sama dengan CE

Analisis ac :

$$R_{12} = R_1 // R_2$$

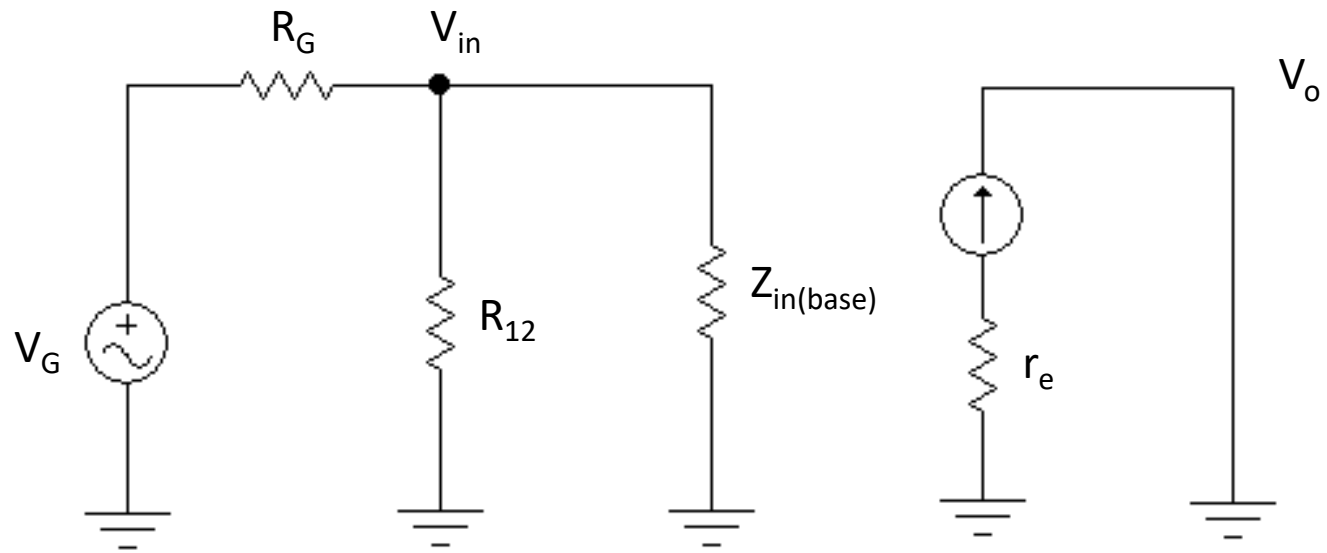
$$r_e = R_E // R_L$$





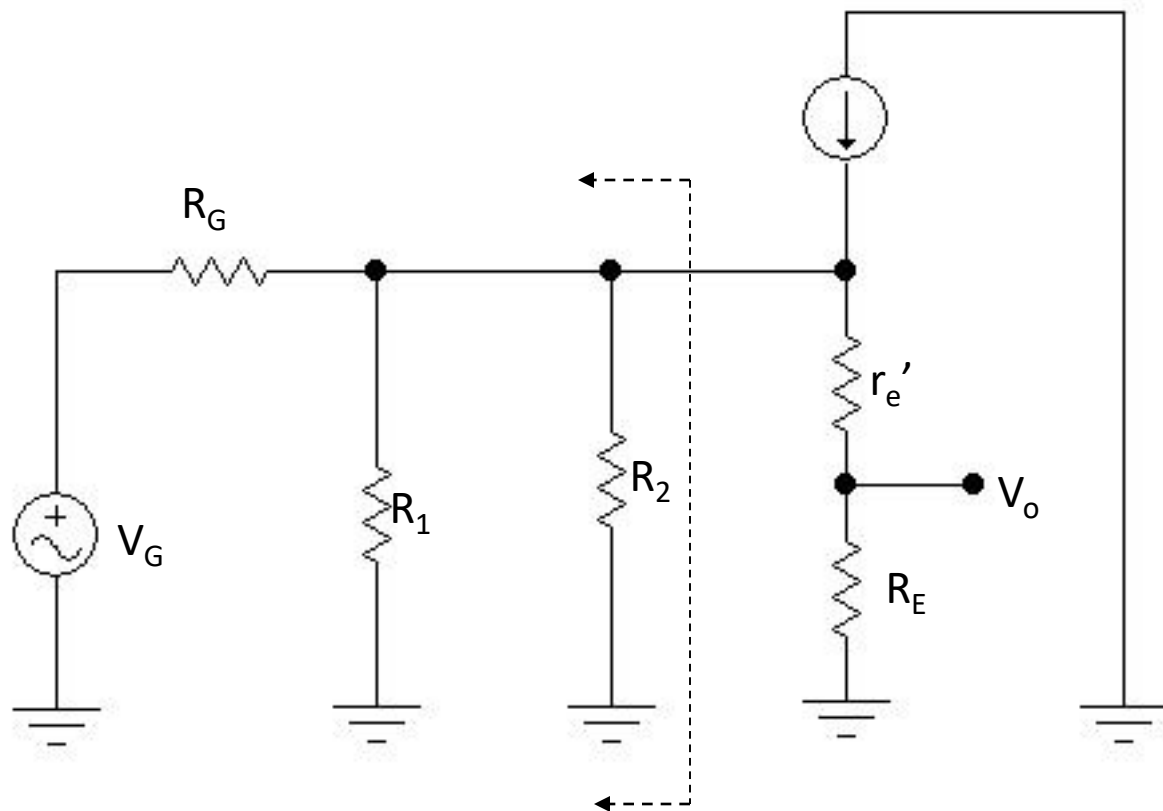
$$v_b = i_e (r_e + r_e') \quad v_o = v_e = r_e i_e \quad \rightarrow \quad A_v = \frac{v_e}{v_b} = \frac{r_e}{r_e + r_e'} < 1$$

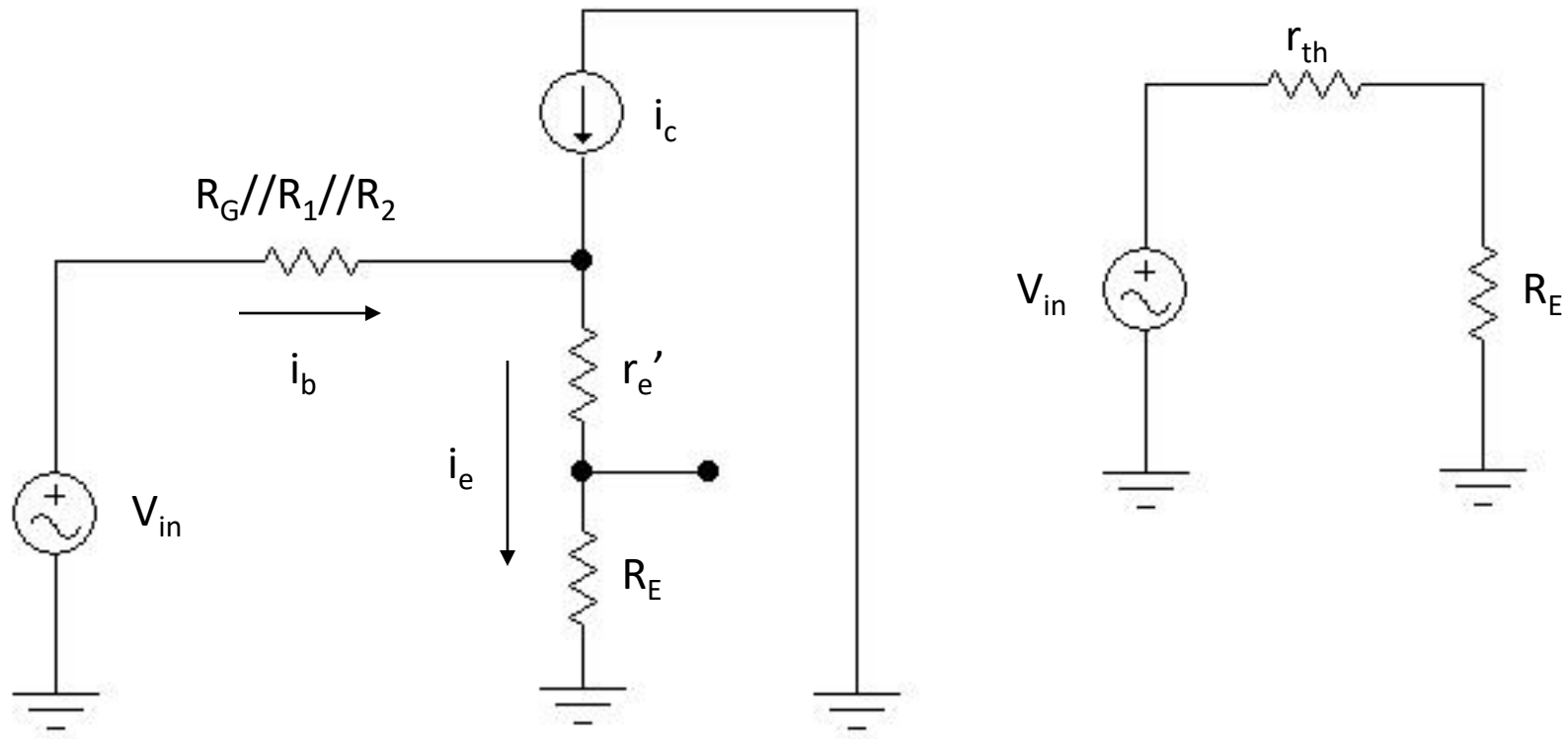
$$Z_{\text{in(base)}} = \frac{v_b}{i_b} = \frac{i_e (r_e + r_e')}{i_b} \approx \beta (r_e + r_e')$$



$$z_{\text{in}} = R_1 // R_2 // \beta(r_e + r_e') \rightarrow v_{\text{in}} = \frac{z_{\text{in}}}{z_{\text{in}} + R_G} V_G$$

$$v_o = A_v v_{\text{in}} = \frac{r_e}{r_e + r_e'} \frac{z_{\text{in}}}{z_{\text{in}} + R_G} V_G$$

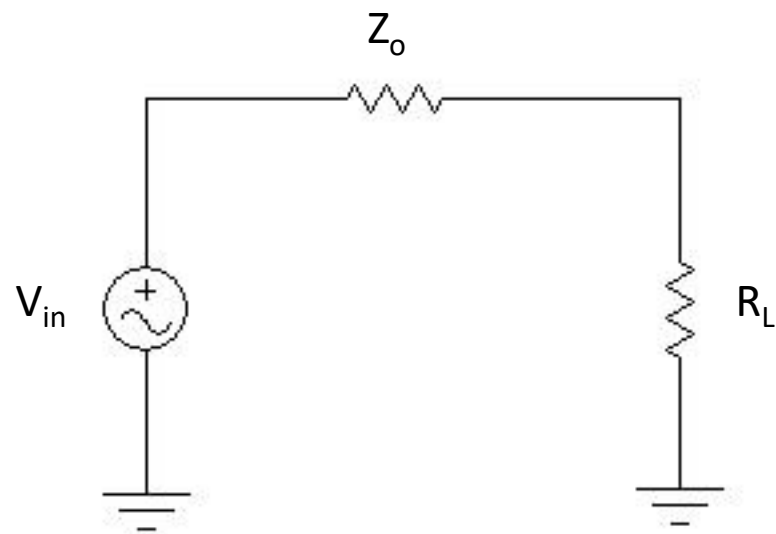
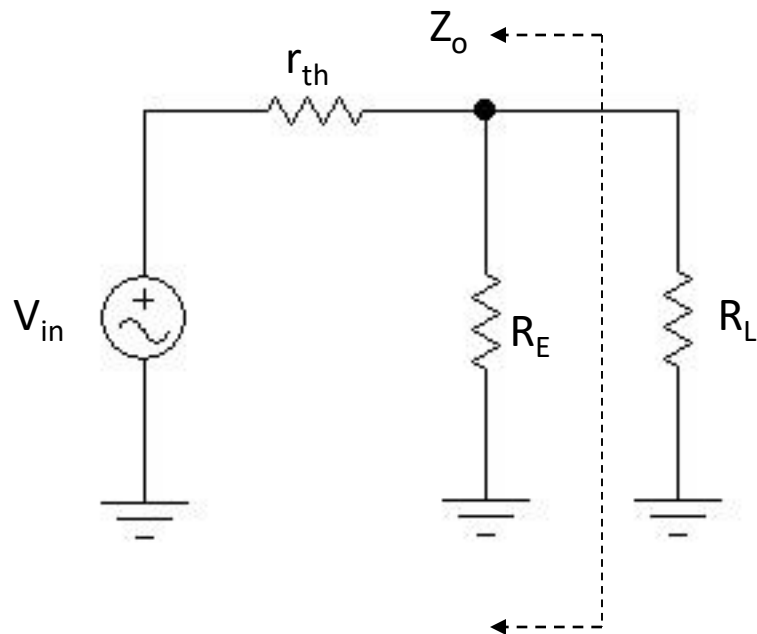




$$r_g = R_G // R_1 // R_2 \rightarrow -V_{in} + i_b r_g + i_e r_e' + i_e R_E = 0$$

$$i_b = \frac{i_c}{\beta} \approx \frac{i_e}{\beta} \rightarrow -V_{in} + (r_e' + R_E + \frac{r_g}{\beta}) i_e = 0$$

$$i_e = \frac{V_{in}}{r_e' + R_E + \frac{r_g}{\beta}} \rightarrow r_{th} = r_e' + \frac{r_g}{\beta} = r_e' + \frac{R_G // R_1 // R_2}{\beta}$$

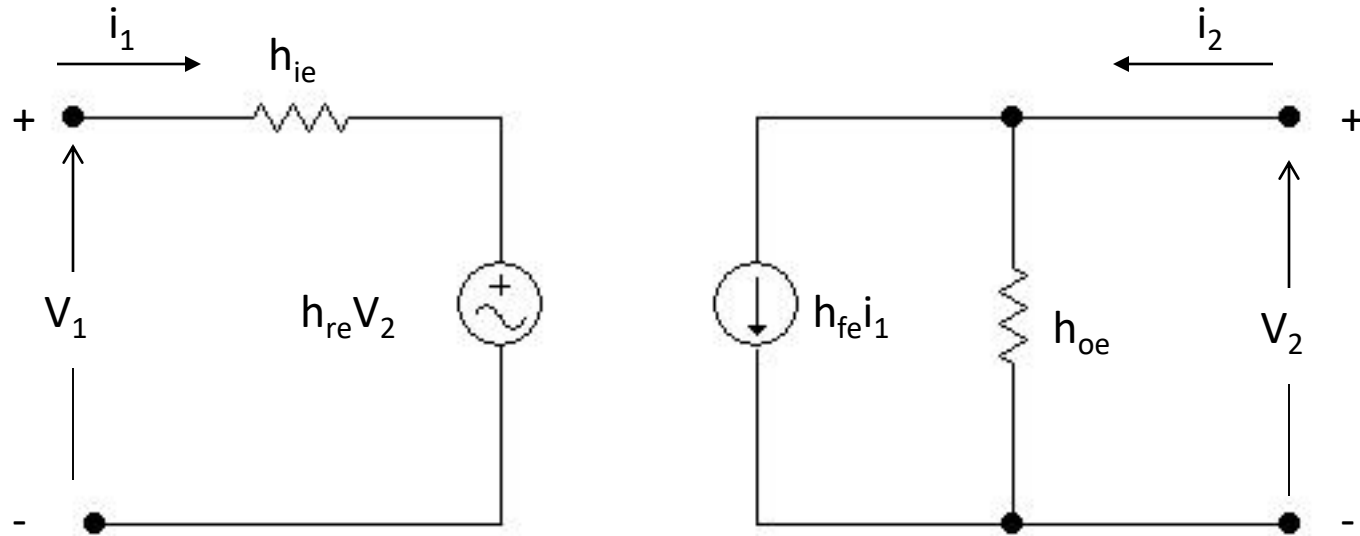


$$Z_o = R_E // \left(r_e' + \frac{R_G // R_1 // R_2}{\beta} \right)$$

$$V_o = \frac{R_L}{R_L + Z_o} V_{in}$$

	COMMON EMITTER	COMMON COLLECTOR	COMMON BASE
A_v	$\frac{r_c}{r_e'}$	$\frac{r_e}{r_e + r_e'}$	$\frac{R_C}{r_e'}$
Z_{in}	$R_1 // R_2 // \beta r_e'$	$R_1 // R_2 // \beta (r_e + r_e')$	r_e'
Z_o	R_C	$R_E // (r_e' + \frac{R_G // R_1 // R_2}{\beta})$	R_C

MODEL HYBRID



$$V_1 = h_{ie} i_1 + h_{re} V_2$$

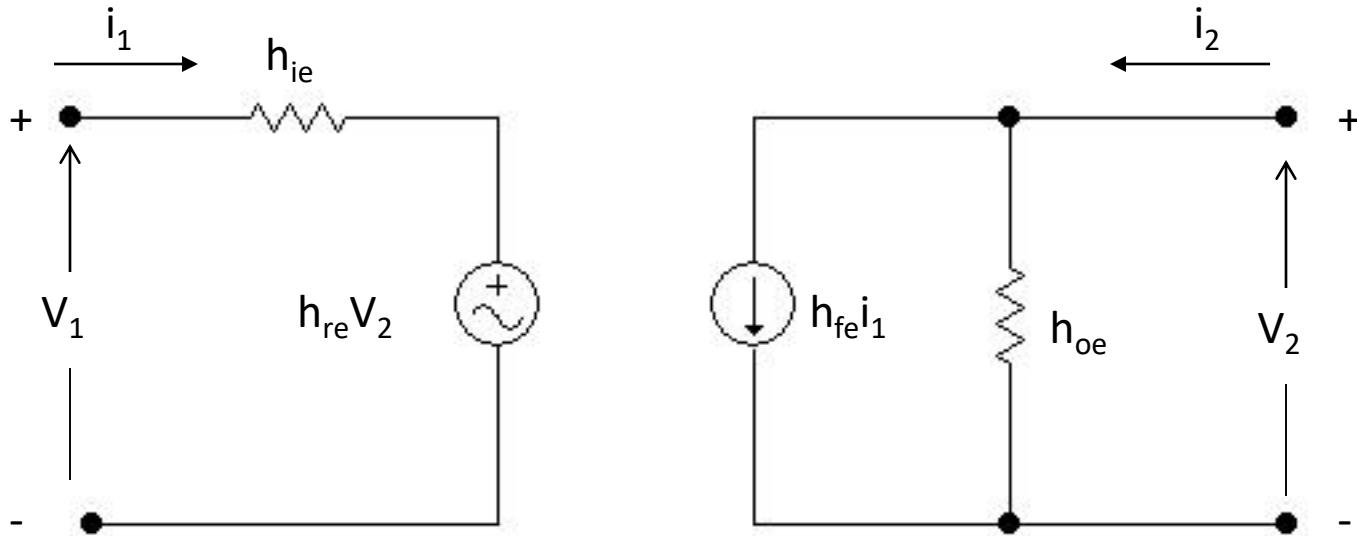
$$i_2 = h_{fe} i_1 + h_{oe} V_2$$

$V_2 = 0$ (output shorted)

$$V_1 = h_{ie} i_1 \rightarrow h_{ie} = \frac{V_1}{i_1} \rightarrow \text{Impedansi input}$$

$$i_2 = h_{fe} i_1 \rightarrow h_{fe} = \frac{i_2}{i_1} \rightarrow \text{Penguatan arus}$$

MODEL HYBRID



$$v_1 = h_{ie} i_1 + h_{re} v_2$$

$$i_2 = h_{fe} i_1 + h_{oe} v_2$$

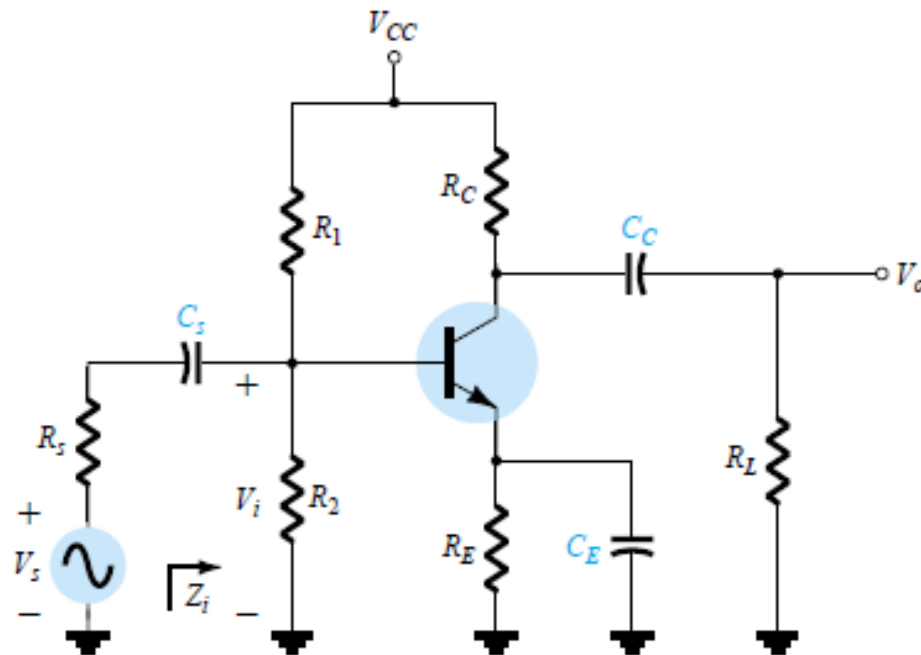
$i_1 = 0$ (input open)

$$v_1 = h_{re} v_2 \rightarrow h_{re} = \frac{v_1}{v_2} \longrightarrow \text{Penguatan tegangan balik}$$

$$i_2 = h_{oe} v_2 \rightarrow h_{oe} = \frac{i_2}{v_2} \longrightarrow \text{Admitansi output}$$

HW

1. Perhatikan rangkaian penguat BJT berikut ini :



Jika diketahui parameter penguat sbb:

$$C_s = 10 \mu\text{F}, \quad C_E = 20 \mu\text{F}, \quad C_C = 1 \mu\text{F}$$

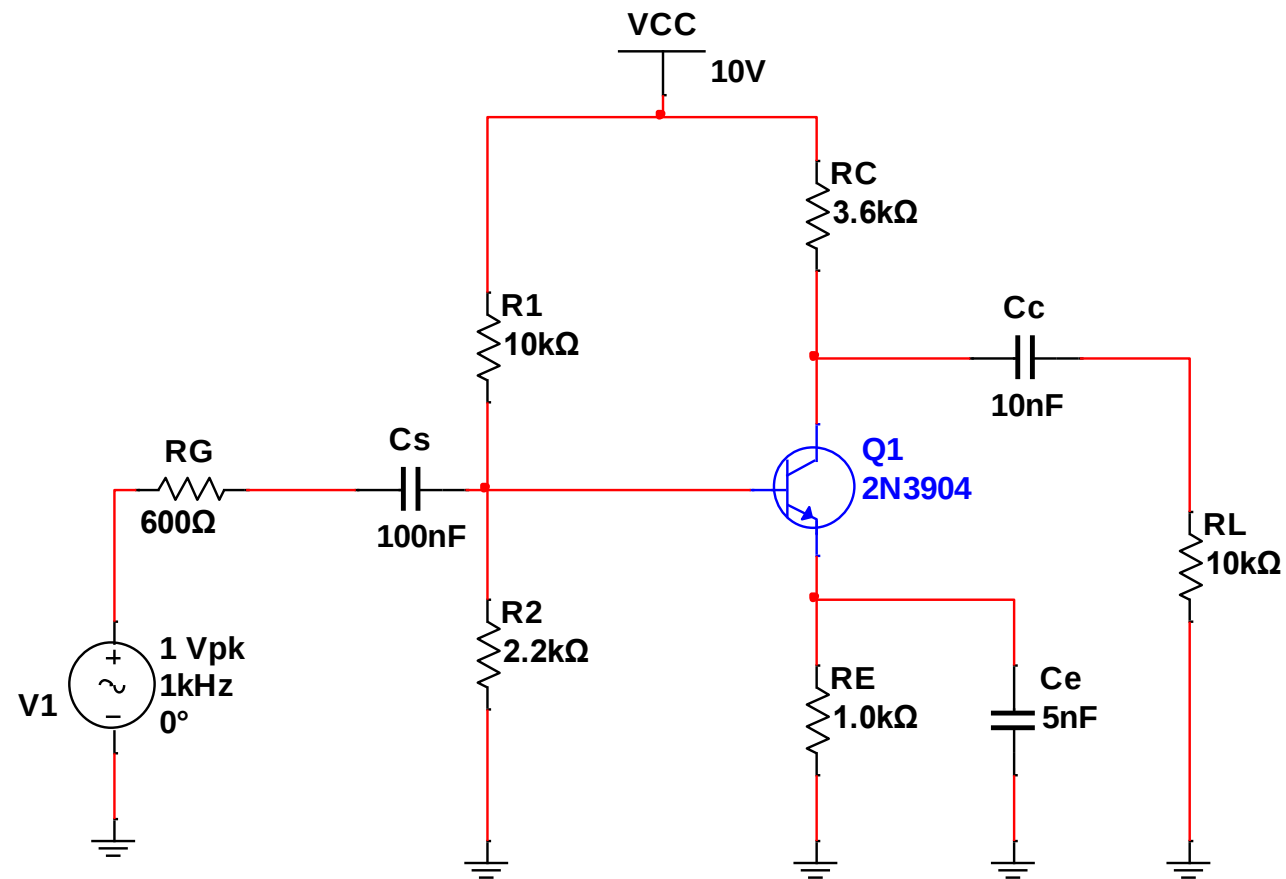
$$R_s = 1 \text{ k}\Omega, \quad R_1 = 40 \text{ k}\Omega, \quad R_2 = 10 \text{ k}\Omega, \quad R_E = 2 \text{ k}\Omega, \quad R_C = 4 \text{ k}\Omega, \\ R_L = 2.2 \text{ k}\Omega$$

$$\beta = 100, \quad r_o = \infty \Omega, \quad V_{CC} = 20 \text{ V}$$

Tentukan respon penguatan pada frekuensi bawah di bagian :

- sisi input-nya !
- sisi output-nya !
- sisi bypass-nya !

2. Simulasikan rangkaian penguat BJT berikut ini dan gambarkan kurva tanggapan frekuensinya



Thank You